

1. PROJECT & TEAM INFORMATION

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models
- **Team ID:** SWTID-2026-7486
- **Team Leader:** Diwan Oli M
- **Team Members:** Siva, Akashdeepan, Abdul Bazith K, Mohamed Ibrahim

2. USER PERSONA DEFINITION

- **Primary User:** Aspiring comic creators, digital storytellers, and students.
- **Objective:** To seamlessly convert a text-based narrative concept into a highly visual, structured comic book format without requiring any manual graphic design intervention.

3. JOURNEY STAGES & TOUCHPOINTS

Stage 1: Platform Onboarding

- **User Action:** The user accesses the ComicCraft web portal via their browser.
- **Touchpoint:** The main landing page (`index.html`).
- **User Emotion:** Intrigued but unsure of the technical complexity.
- **System Opportunity:** Display a clean, minimalist UI with clear input fields and dropdowns (e.g., Art Style, Tone) so the user feels guided and confident.

Stage 2: Data Input & Submission

- **User Action:** The user enters the core story prompt, names the protagonist, selects a specific setting, and clicks "Generate Comic".
- **Touchpoint:** Frontend Form to FastAPI POST endpoint.
- **User Emotion:** Enthusiastic and eager to see the results.
- **Pain Point:** AI processing can take time; users might think the page is frozen.
- **System Opportunity:** Integrate a dynamic loading indicator to reassure the user that the AI models are actively processing their request.

Stage 3: Background Orchestration (System Side)

- **System Action:** The backend securely communicates with Google Gemini for the storyline and Hugging Face Stable Diffusion for the panel illustrations.
- **Touchpoint:** External AI API connections.
- **System Goal:** Maintain stable connections and execute failover mechanisms if any external AI server experiences high traffic.

Stage 4: Output Review

- **User Action:** The user is presented with the generated comic strip layout on their screen.
- **Touchpoint:** The interactive preview interface (`comic_preview.html`).
- **User Emotion:** Surprised and delighted by the visual translation of their words.
- **System Opportunity:** Ensure the UI clearly separates the generated image, panel title, and narrative text for optimal readability.

Stage 5: Final Export

- **User Action:** The user initiates the PDF download to save their creation locally.
- **Touchpoint:** The export success route and file download prompt.
- **User Emotion:** Highly satisfied and ready to share their work.
- **System Opportunity:** Generate a perfectly aligned, multi-page PDF document using the FPDF library.